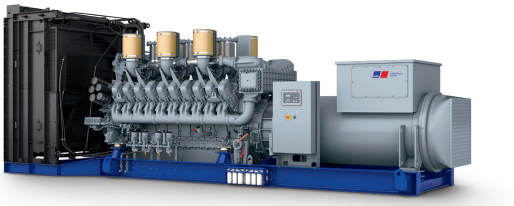
Diesel Generator Set

mtu 20V4000 DS3100

380V – 11 kV/50 Hz/grid stability power/ NEA (ORDE) optimized/20V4000G24F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 2800 kVA - 2910 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- NEA (ORDE) optimized

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data[1]

Engine

Manufacturer	*mtu*
Model	20V4000G24F
Type	4-cycle
Arrangement	20V
Displacement: l	95.4
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ECU 9
Max power: kWm	2420
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	27

Fuel consumption[2]

	l/hr	g/kwh
At 100% of power rating:	574.4	197
At 75% of power rating:	450.5	206
At 50% of power rating:	319.3	219

Liquid capacity (lubrication)

Total oil system capacity: l	390
Engine jacket water capacity: l	205
Intercooler coolant capacity: l	50

Combustion air requirements

Combustion air volume: m^3/s	2.7
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m^3/hr	80
Coolant flow rate (LT circuit): m^3/hr	32.5
Heat rejection to coolant: kW	980
Heat radiated to charge air cooling: kW	410
Heat radiated to ambient: kW	105
Fan power for electr. radiator (40°C): kW	70

Exhaust system

Exhaust gas temp. (after turbocharger): °C	550
Exhaust gas volume: m^3/s	7.1
Maximum allowable back pressure: mbar	85
Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA53.2 M9 (Low voltage Leroy Somer standard)	380 V	2320	2900	4406	2240	2800	4254
	400 V	2320	2900	4186	2240	2800	4041
	415 V	2320	2900	4034	2240	2800	3895
Marathon 1030FDL7094 (Low voltage Marathon)	380 V	2320	2900	4406	2256	2820	4285
	400 V	2320	2900	4186	2256	2820	4070
	415 V	2320	2900	4034	2256	2820	3923
Marathon 1030FDH7101 (Medium volt. marathon)	11 kV	2320	2900	152	2256	2820	148
Leroy Somer LSA53.2 ZL14 (Medium volt. Leroy Somer)	11 kV	2328	2910	153	2264	2830	149

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.